Design of optical metamaterial mirror with metallic nanoparticles for floating-gate graphene optoelectronic devices

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Abstract: The purpose of this work is to conceive the idea for using the gate dielectrics of floating-gate memory device (i.e., Au nanoparticle (AuNP) monolayer embedded within polymeric matrix) as a magnetic mirror, so as to harness the broadband light absorption of thin film optoelectronics. In particular, we systematically examined whether the versatile assembly of spherical AuNP monolayer can be indeed treated as the effective magnetic mirror for floating-gate graphene optoelectronic device. High amenability of the AuNP assembly with the large-area device fabrication procedures may make this strategy widely applicable to various thin film optoelectronic devices. Our study thereby advances the design of mirror for thin film optoelectronics.

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1. Introduction

In thin film optoelectronic technologies (e.g., photodetectors and solar cells), there has usually been a trade-off between efficient photon absorption and the enhancement of electrical performance [1–3]. In particular, the reduction of active film thickness represents the increase in the efficiency of charge/carrier collection, while it is inherently accompanied by sacrificing light absorption. This trade-off problem becomes more prominent, when the two-dimensional (2D) materials such as graphene and molybdenum disulfide (MoS_2) are implemented into optoelectronic devices as an active layer. Even if the exotic electronic properties of 2D materials (i.e., high charge carrier mobility even at room temperature) promise compelling opportunities in advancing various optoelectronics [4], their poor optical cross-section remains as obstacles for high performance device operation [5].

Metallic backplane mirror (i.e., just flat metallic film) has been a popular component in thin film optoelectronic devices, as allowing us broadband photon recycling [6,7]. However, the translation of a flat metallic mirror into the thin film optoelectronic devices still requires optical engineer to address the problem of significantly reduced $|E|$ within one-quarter of wavelength (i.e., an electric mirror) [8]. Very recently, Esfandyarpour et al. [8], has promoted this issue by using electron-beam lithographic texturing of metallic surface (a magnetic mirror), but the difficulty in large-area fabrication together with relatively low compatibility with various optoelectronic devices have stymied its practical applications. Herein, we propose a new design of a metamaterial mirror, in which a versatile assembly of large-area gold nanoparticles (AuNPs) array within a dielectric matrix (acting as a gate dielectrics for charge trapping in floating-gate transistor) can result in the maximized $|E|$ right at mirror surface (a magnetic mirror). In particular, we theoretically verify how a AuNP monolayer can achieve a magnetic mirror by the analysis on the collective set of optical properties including band structure, impedance, reflection and $|E|/|H|$ spatial distribution. Also, high amenability of the AuNP assembly with the large-area device fabrication may make this strategy widely applicable to various thin film optoelectronics. Our study thereby advances the design of mirror for rational engineering of light-matter interaction within thin film optoelectronics.

2. Design of optical metamaterial mirror

2.1 AuNP array as a magnetic mirror (metamaterial mirror)

We qualitatively outline how high surface impedance of spherical AuNP monolayer (hexagonally closed-packed array, see the inset of Fig. 1(a)) can be harnessed to realize magnetic mirror (a minimized phase reversal of reflection light). Much like other periodically bumpy metallic surfaces [8–10], AuNP array within dielectric matrix (i.e., poly-4-vinylphenol, cPVP) can efficiently make the path of surface current less straightforward, so as to effectively reduce optical conductivity (σ). This reduced σ in turn results in the enhancement of E_z at the surfaces of metallic structure [10]. In our structure, both surface plasmon polaritons (SPPs) and photonic (PhC) modes can lead to the dramatic reduction of σ

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(enhancement of E_z). As the impedance (Z) at the surface of metallic structures is given by the E_z/H_y, the increase in E_z at the surfaces leads to the reduction of the reflection according to following complex reflection coefficient [8,11]:

$$r = r_0 e^{i\varphi} = \frac{Z_{S1}\cos(\theta_i) - Z_{S2}\cos(\theta_t)}{Z_{S1}\cos(\theta_i) + Z_{S2}\cos(\theta_t)} \quad (1)$$

where Z_{S1}, Z_{S2}, φ, θ_i, θ_t, and r_0 denote the impedance at the metal surface, the wave impedance of the incident medium, reflection phase, incident angle, transmitted angle, and reflection amplitude, respectively. For the normal irradiation, second terms of the numerator and dominator of Eq. (1) become unity; then, the magnitude of reflection is simplified by Z_{S1} (or σ) of the structure ($r = (Z_{S1}-1)/(Z_{S1}+1)$). Thus, the increase in Z_{S1} via SPPs or PhC modes allows us to minimize the phase reversal.

2.2 Properties of metamaterial mirror

To confirm above intuitive insight into AuNP array, we have carried out numerical calculation by finite-difference, time-domain (FDTD) simulation (see detailed descriptions about FDTD simulation in Appendix section). In this study, we used 150 nm sized spherical AuNP as a structural primitive. Also, we assumed that AuNP array is embedded into 160 nm thick cPVP matrix. This is mainly due to the fact that the encapsulation of AuNP array with polymeric matrix can facilitate the implementation of this structure into the device [12].

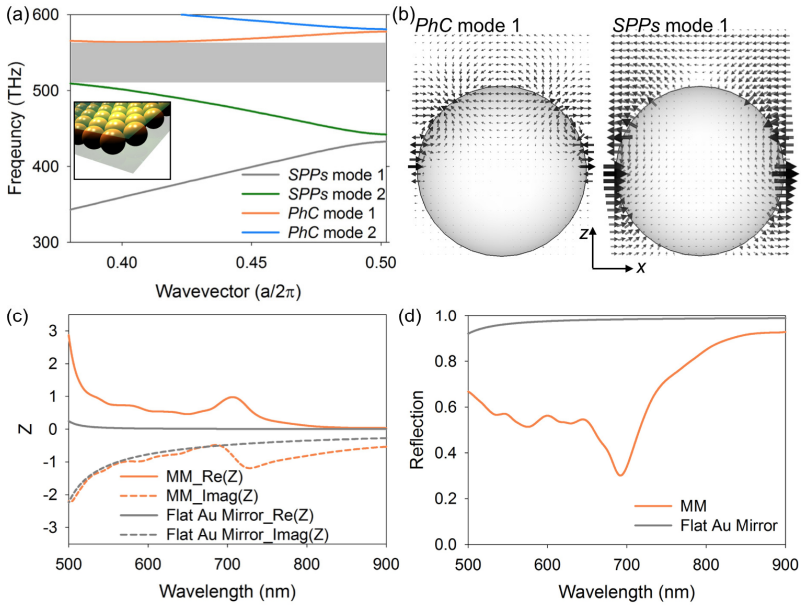


Fig. 1. (a) Band structure of AuNP array, embedded within 160 nm thick cPVP matrix. (b) The spatial distributions of E_z vector for PhC mode 1 and SPPs mode 1 at k = a/2π. (c) Retrieved impedance and (d) simulated reflection of AuNP array (metamaterial mirror, MM) and a flat Au.

Figure 1 presents the collective set of optical properties of the designed metamaterial mirror layer including band structure, E_z distribution at the bandedge of $k_x a/2\pi$, impedance, and reflection. The spatial distribution of E_z (Fig. 1(b)) reveals the nature of optical modes of the structures. Particularly, the E_z of PhC mode 1 is found to be mostly oriented along the

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horizontal direction, whereas the spatial distributions of E_z are vertically oriented for the case of SPPs mode 1 [13]. PhC mode 2 and SPP mode 2 exhibit similar E_z distributions, compared with PhC mode 1 and SPPs mode 1, respectively. The plasmonic and photonic bandgaps are opened at the wavelength of 707 nm (424 THz) and 520 nm (577 THz), respectively; another forbidden region (see grey colored box in Fig. 1(a)) is also observed (photonic modes).

Next, we numerically retrieved the effective impedance (Z). As the size of individual AuNP is much smaller compared to the wavelength of interest, the applications of both homogenization theory and effective parameter retrieving method (using the scattering parameters) can be justified [14]. Note that the peak of impedance was observed near at the forbidden regions such as plasmonic and photonic bandgaps (Fig. 1(c)). Also, the enhancement of impedance can be achieved at the broadband visible frequency. The huge enhancement of impedance near at 500 nm is due to the interband transition of Au rather than reflection phase control. According to the complex reflection coefficient (Eq. (1)), the reflection amplitude (Fig. 1(d)) inversely follows the trend of impedance changes. Then, we compared the impedance and reflection of AuNP array with those of a flat Au (150 nm thick flat Au embedded within 160 nm thick cPVP). Obviously, a very low impedance (less than 0.1) of a flat Au layer results in the almost 100% reflection in stark contrast to the high impedance of AuNP array.

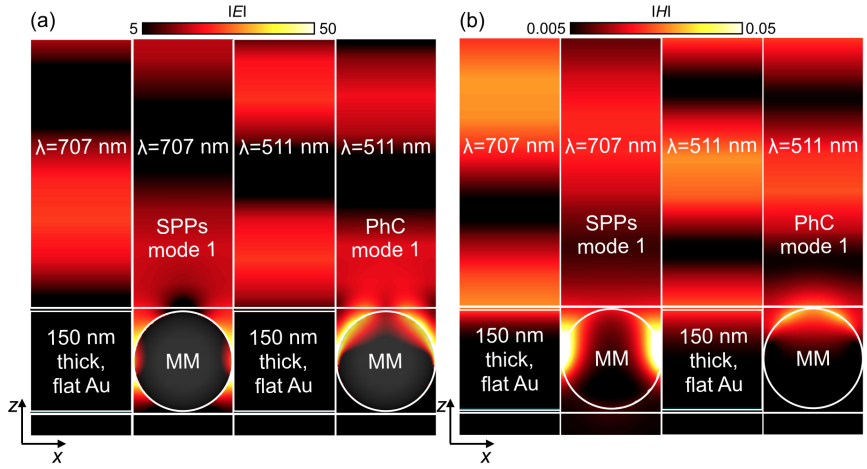


Fig. 2. (a) $|E|$ and (b) $|H|$ distributions for a flat Au layer and AuNP array (metamaterial mirror, MM) at wavelength of 707 nm (SPPs mode 1) and 511 nm (PhC mode 1).

For analysis in more depth, we characterized $|E|/|H|$ field distributions (Fig. 2) and the reflection phase. These studies were carried out at the wavelength of 511 nm (photonic bandgap) and 707 nm (plasmonic bandgap). The light reflecting off a flat Au layer shows a standing wave with a minimized $|E|$ near at its surface, resulting from in-phase superposition between incident and phase reversed ($\varphi = \pi$) reflecting light.

In contrast, AuNP array can exhibit far more $|E|$ right at its surface. This is because the phase shift of the light reflecting off AuNP array can be reconfigured to be much smaller than π (i.e., $\varphi = \pi/6.3$ at wavelength of 707 nm and $\varphi = \pi/7.1$ at wavelength of 511 nm). This reconfigured reflection phase through SPPs and PhC modes is well revealed by the shifted sinusoidal envelop of standing wave with respect to $|E|$ profile of a flat Au mirror. It is also worth noting that overall $|E|$ of standing wave above AuNP array is weaker than that above a flat Au. This further evidences both the stored incident light and accumulated phase into AuNP array.

Analyzing $|H|$ distribution near at the surface of mirror also supports such analysis. The $|H|$ near at the surface of AuNP array is found be weaker than that near at a flat Au. As such,

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AuNP array can harness higher impedance and thus more concentrated |*E*| near at its surface, compared to a flat Au.

2.3 Flexibility in tuning mirror properties

The additional controllability of metamaterial mirror properties can accrue, if we use spherical AuNPs as a building block. For example, AuNPs can exist as various structural motifs, such as core-shell and metallic alloy architecture [15,16]. Thereby, we can obtain additional flexibility in the control of SPP or PhC modes and thus reflection behavior of metamaterial mirror. Toward this direction, we further expanded the available range of metamaterial mirror properties by using silica core (120 nm in diameter) and Au shell (15 nm in thickness) NP array embedded within cPVP matrix (see the inset of Fig. 3(a)).

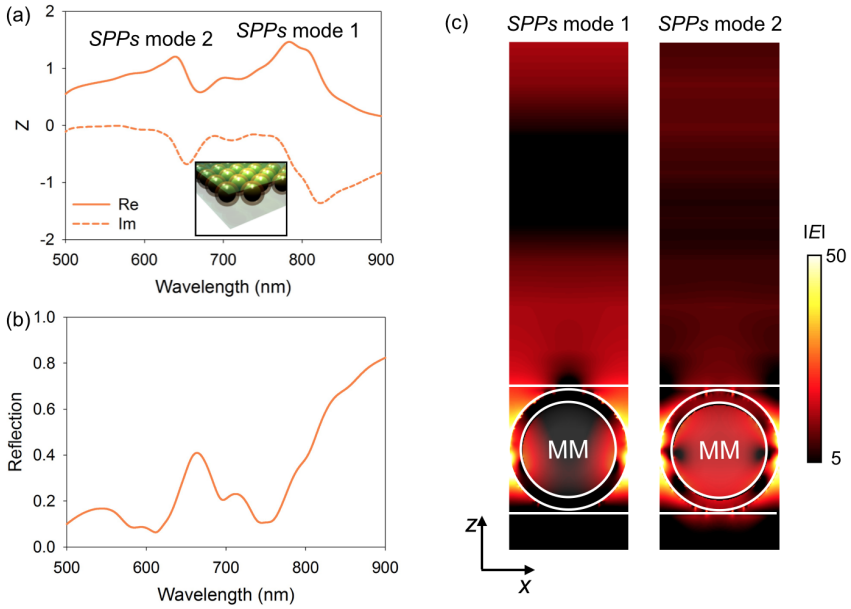


Fig. 3. (a) Retrieved impedance and (b) simulated reflection of silica-Au core-shell NP array. (c) |*E*| distribution of silica-Au core-shell NP array (metamaterial mirror, MM) at SPP mode 1 and mode 2.

For silica-Au core-shell NP array, the photonic modes disappear at the wavelength of interest mainly due to the negligible difference of refractive index between silica and cPVP. Also, the resonant wavelength of the fundamental SPP mode is shifted to 790 nm from 707 nm by inserting silica core into AuNPs. Meanwhile, the core/shell geometry makes the conduction path more tortuous, so as to further increase the impedance and reduce the reflection, compared to simple AuNPs (e.g., 1.25 at SPPs mode 2 and 1.49 at SPPs mode 1).

Moreover, at the wavelength of SPPs modes, silica-Au core-shell NP array exhibits weaker intensity of standing wave, compared with AuNP array (compare Fig. 3(c) with Fig. 2(a)). This is another evidence of more accumulated phase ($\varphi = \pi/7.45$ at wavelength of 610 nm and $\varphi = \pi/7.61$ at wavelength of 790 nm). Nonetheless, as |*E*| is mainly concentrated at the vicinity of core-shell NPs (i.e., absorption within NPs array) rather than at the surface of mirror, silica-Au core-shell NPs array shows less favorable spatial distribution of |*E*| in terms of thin film optoelectronic applications. These results imply that on-demand control of |*E*| distribution as well as impedance and reflection phase has to be carried out for designing the mirror of thin film optoelectronic devices.

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3. Device application

Herein, we chose the pentacene-graphene nano-floating gate transistor memory (NFGTM) as a model optoelectronic device. NFGTM based on gate dielectrics (charge trapping layer) has been treated as a promising nonvolatile memory owing to excellent reliability and capability of multilevel programming [12]. Recently, we had identified pentacene-coated graphene NFGTM as a non-volatile photodetector with high optical cross-section [12]. The device architecture (Fig. 4(a)) is as follows (from substrate to active channel layer): plastic substrate (polyethylene naphthalate, PEN), a 300 nm thick indium tin oxide (ITO) back-gate electrode, 30 nm thick aluminum oxide blocking dielectric layer (i.e., amorphous Al_2O_3 with 30% oxide and 70% aluminum), AuNP array within 160 nm thick cPVP matrix (i.e., gate dielectrics for charge trapping), and 25 nm thick pentacene-graphene hybrid active channel layer. The rationally designed gate dielectrics with AuNP array can act as a metamaterial mirror for the enhancement of broadband light absorption within active channel layer. Especially, to take advantage of memory functionality (i.e., trapping charges within AuNPs via tunneling effect), this metamaterial mirror should be placed right below the active channel layer [12], as shown in Fig. 4(a) and Appendix section.

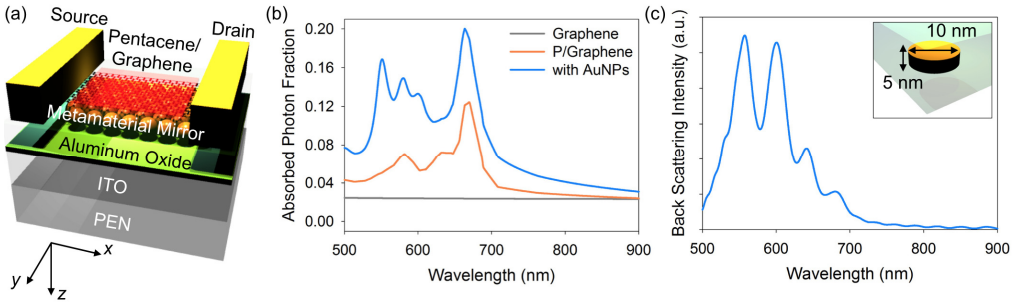


Fig. 4. (a) Schematic for pentacene-graphene nano-floating gate transistor memory (NFGTM) with metamaterial mirror. (b) Absorbed photon fraction of graphene, 25 nm thick pentacene/graphene hybrid, and 25 nm thick pentacene/graphene incorporated with Au nanodisc. (c) Backward scattering spectra of the disc-shaped AuNP (10 nm in diameter and 5 nm in thickness).

3.1 Optical properties of NFGTM

As presented in Fig. 4(b), graphene itself generally shows poor optical cross-section at visible frequency of interest (absorption of 2.3%). Only recently, have we found that such low optical cross-section of graphene can be effectively addressed by coating pentacene on it [12]. Especially, the efficient generation of electron-hole pair within pentacene and subsequent transferring of the generated electron-hole pair to graphene via capacitive coupling was found to dramatically increase the photo-responsibility of the graphene optoelectronic devices [12]. More importantly, the presence of AuNPs within gate dielectrics can give rise to the plasmonic back scattering-enabled enhancement of light absorption as well as tunneling-mediated charge trapping [12]. Indeed, as presented in Figs. 4(b)-4(c), the absorption of active channel layer can be further enhanced at the plasmonic scattering wavelength of AuNPs (the disc-shaped AuNP with 5 nm height and 10 nm diameters, embedded in 20 nm thick cPVP dielectric matrix). This success to the increase in device absorption by means of plasmonic light scattering inspires the current work on the rational design of gate dielectrics to be useful for a magnetic mirror, in that provides a way for recycling broadband light within NFGTM.

3.2 Enhancement of broadband light absorption in NFGTM

We finally profiled the enhancement of broadband light absorption in active channel layer by using the designed metamaterial mirror. The numerical simulation was performed with full

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device architecture (see Appendix section). Figure 5 compares the absorbed photon fraction of active channel layer for the absence of metallic mirror, a flat Au mirror, AuNP mirror, and silica-Au core-shell NP mirror.

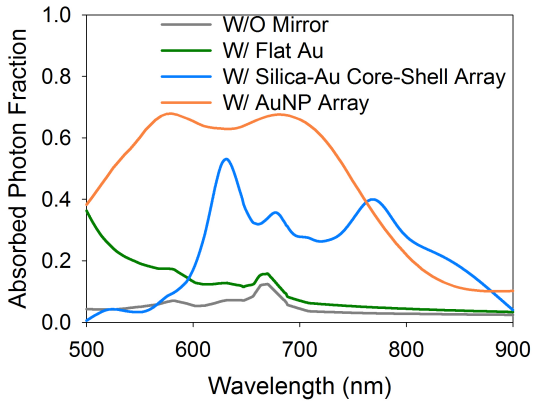


Fig. 5. Photon absorption fraction within active channel layer for different mirror designs.

A flat Au mirror can still provide the broadband absorption enhancement of active channel layer. In our case, the thickness of active channel layer (~25 nm) is much smaller than the wavelength of interest. Compared with highly absorbing germanium [7], the absorption of pentacene is relatively moderate. Thus, its phase accumulation through round trip propagation (i.e., optical resonance) should make a negligible contribution to the enhancement of broadband light absorption. This enhancement of broadband light absorption simply originates from the increased overlap between pentacene layer and $|E|$ distribution. In particular, as the wavelength is reduced, the absorbed photon fraction becomes more significant due to the interband transition-enabled energy storage and reflection phase accumulation at the surface of a flat Au. This aspect is well reflected by the series of $|E|$ spatial distributions at different wavelength (Fig. 6(a)). We can also observe that $|E|$ of standing wave above a flat Au at wavelength of 511 nm is further reduced after the implementation of active channel layer into the device (compare the third panel of Fig. 2(a) with first panel of Fig. 6(a)).

Even if φ is a bit larger than zero, the use of AuNP array can result in far more enhancement of the broadband light absorption (Fig. 5). Fundamental to this dramatic enhancement of broadband light absorption is detailed in series of $|E|$ spatial distribution (Fig. 6(b)). In particular, the enhancement of broadband light absorption is dominantly in regards to the highly concentrated $|E|$ near at the surface of AuNP array. This is main reason that PhC modes can be advantageous over the SPPs modes for broadband light absorption of active channel layer. Such further enhanced light absorption within active channel layer also results in the reduced intensity of standing wave above the AuNP array compared to that above a flat Au (compare Figs. 6(b) with 6(a)).

In stark contrast, silica-Au core-shell NP array strongly absorbs the incident light within each individual NP through SPP whispering gallery-like modes (e.g., 511 nm) (see Fig. 6(c)). Thus, this mirror likely accumulates $|E|$ at the vicinity of silica-Au core-shell NPs rather than at its surface; consequently, being less favorable for broadband light absorption in active channel layer. Indeed, the absorbed photon fraction of active channel layer gets reduced after inserting silica core into AuNPs (see Fig. 5).

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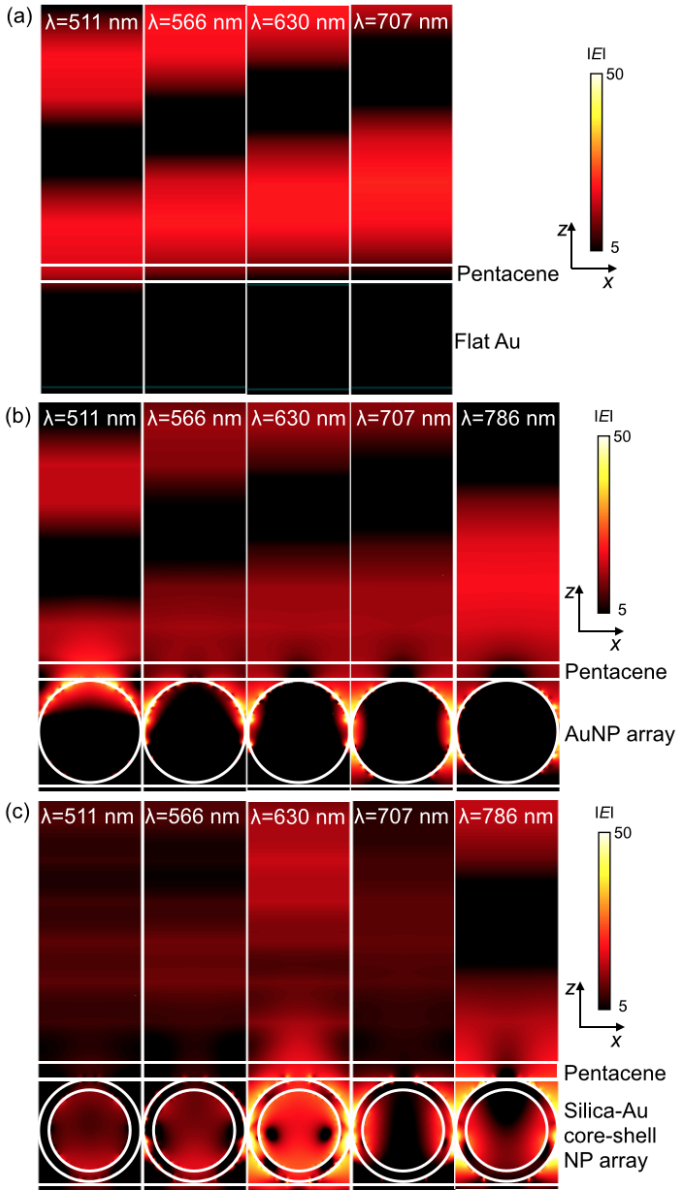


Fig. 6. (a-c) |*E*| distribution of the NFGTM with a different mirror design: (a) flat Au mirror, (b) AuNP array, and (c) silica-Au core-shell NP array.

4. Conclusions

Here, we have suggested theoretical designs of metamaterial mirror for achieving the enhancement of broadband light absorption in thin film optoelectronic device. Especially, we have revealed that versatile assembly of AuNP monolayer allows us to achieve an effective magnetic mirror (i.e., SPP and PhC mode-enabled high impedance and the minimized phase reversal). With these advances, a more flexibility in the design of thin film optoelectronic devices should now become available for both rational management of photon flow and the enhancement of device performance.

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5. Appendix

5.1 FDTD

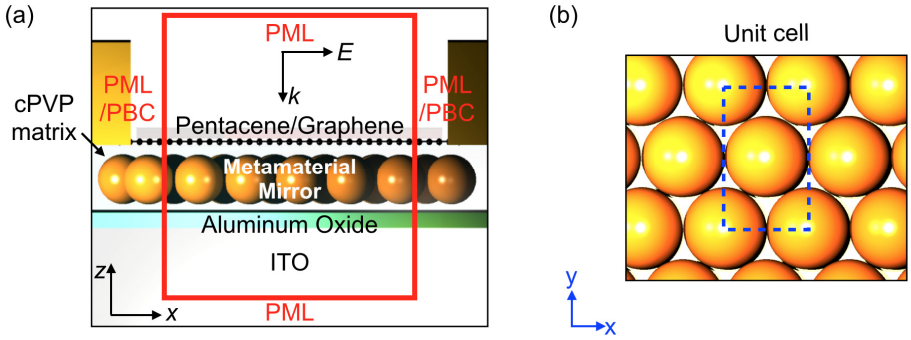


Fig. 7. (a) The 3D full field electromagnetic simulation geometry by finite-difference, time-domain (FDTD) method. (b) Top view of unit cell structure, dotted by blue dotted box.

We carried out 3D full field electromagnetic simulation by home-built FDTD code with the boundary conditions detailed in Fig. 7(a). The red colored box highlights perfectly matched layer (PML) and periodic boundary conditions (PBC). AuNPs and silica-Au core-shell NPs were modeled to be encapsulated by 1 nm organic surfactant (poly(diallyl dimethyl ammonium chloride), polyDADMAC) [15]. In order to avoid unrealistic point contact, the interface between the hexagonally close-packed AuNPs was truncated by 1 nm. Unit cell is also indicated in Fig. 7(b) (blue dotted box). Due to the hexagonal geometry, the optical properties are independent on the polarization of normally incident light. For the calculation of the complex permittivity of Au, Drude-critical model was implemented into FDTD code [15]. The complex permittivity of other materials including PEN, ITO, polyDADMAC, and pentacene were experimentally measured by ellipsometry. The Bruggeman effective medium approximation allows us to obtain the complex permittivity of Al_2O_3 [17]:

$$n_{al}\frac{(\varepsilon_{Al}-\varepsilon)}{(\varepsilon_{Al}+2\varepsilon)}+n_{oxide}\left(\frac{\varepsilon_{oxide}-\varepsilon}{\varepsilon_{oxide}+2\varepsilon}\right)=0 \qquad (2)$$

where n_{al} and n_{oxide} are the volume ratios of Al and oxide, respectively; ε is the permittivity of amorphous Al/Al_2O_3 composite. The both ε_{Al} and ε_{oxide} were obtained by Drude model and empirical measurement, respectively. To calculate the absorbed photon fraction within pentacene/graphene hybrid layer (i.e., active channel layer), $|E|$ was integrated over one period of unit cell. FDTD code employed subcell averaging and simulation was continued until a state of convergence reached.

5.2 Graphene simulation

For graphene, Fermi-level variable complex optical conductivity, $\sigma(\omega) = \sigma_{intra}(\omega) + \sigma_{inter}(\omega)$, was first obtained by means of Kubo formula [18]:

$$\sigma_{inter}(\omega)=i\frac{e^2\omega}{\pi}\int_{\Delta}^{\infty}d\epsilon\frac{(1+\frac{\Delta^2}{\epsilon^2})}{(2\epsilon)^2-(\hbar\omega+i\Gamma)2}\times\left[f(\epsilon-E_F)+f(\epsilon+E_F)\right] \qquad (3)$$

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$$\sigma_{intra}(\omega) = \frac{e^2}{\pi\hbar^2}\frac{i}{\omega + i\tau^{-1}}\int_{\Delta}^{\infty} d\epsilon \left(1 + \frac{\Delta^2}{\epsilon^2}\right) + \left[f(\epsilon - E_F) + f(\epsilon + E_F)\right] \quad (4)$$

where $f(\epsilon - E_F)$ is the Fermi distribution function with Fermi energy (E_F), Γ indicates the broadening of the interband transitions, τ corresponds to the momentum relaxation time (herein, 250 fs was employed) caused by carrier intraband scattering, and Δ is a half bandgap energy from the tight-binding Hamiltonian near *K* points of the Brillouin zone.

Then, this calculated optical conductivity was translated into the *effective* complex permittivity, as detailed in the reference [19]. The thickness of graphene was set to be 1 nm (i.e., volumetric graphene layer with finite thickness, also referred to as volumetric permittivity approach) and the corresponding permittivity was assigned. However, this chosen thickness of graphene is not essential, since much smaller than the wavelength of interest. We arbitrary varied the thickness of graphene between 0.5 nm to 1.0 nm and matched the corresponding permittivity accordingly. According to the thickness of graphene, the results were not changed significantly. The uniaxial anisotropic permittivity tensor was introduced into our FDTD simulation (the in-plane and out-of-plane components of the permittivity tensors were employed).

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